An Exact Approach to Solving the Drop the Handkerchief Game

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Abstract

Drop the Handkerchief (DTH) is a death game in the final arc "Surpassing The Leader" of the manga *Usogui* by Toshio Sako. It is a two-player, zero-sum stochastic game with simultaneous actions and full observation. We solve the complete 60-action timing model under the frozen survival law of Section 2. After a player can no longer survive an injection, the game does not read their cumulative time dead (TTD) again. A quotient by that fact collapses the 5.27×10^9 states to 289,374,121 classes. A potential function puts the classes into 1,201 layers, and one backward sweep solves them.

Each value has a mixed-strategy gap of at most 10^{-6} against its full 60×60 stage matrix. The matrix is triangular Toeplitz, so one 59-step recurrence gives both equilibrium strategies. One source file builds the whole table in 49 seconds on a fifteen-core laptop. At the root state $V = 0.08985 \pm 0.00061$, so the first Dropper wins with probability 0.54493 ± 0.00031 under optimal play.

1 Introduction

DTH is the pure timing core of the Surpassing the Leader arc wager in *Usogui*. Two players alternate as *Dropper* and *Checker*. In each round, both players choose a second in $\{1, \ldots, 60\}$. A successful check adds the inclusive delay to the Checker's squandered time (ST), and then the roles swap. A failed check injects that ST plus sixty seconds. The Checker revives with the probability in Figure 1, or loses.

The game is finite but large, and each round can need a mixed strategy. The solution has four steps:

1. quotient the profiles for which failure is certainly fatal (Section 4);
2. rank every transition with a potential that always increases (Section 5);
3. solve each 60×60 stage game from its 61 different entries and keep a gap certificate (Sections 3 and 6);
4. run the sweep as one fused kernel over each layer (Section 7).

At a state x, each player chooses a distribution over the sixty seconds:

$$\Delta_{60} = \left\{ \sigma \in \mathbb{R}^{60} : \sigma_i \geq 0,\ \sum_{i=1}^{60} \sigma_i = 1 \right\}.$$

*Code and instructions to reproduce the results are available in the project repository.

The stage matrix $M(x)$ gives the payoff of each action pair, so

$$V(x) = \max_{\sigma_d \in \Delta_{60}} \min_{\sigma_c \in \Delta_{60}} \sigma_d^{\mathsf{T}} M(x) \sigma_c.$$

Each entry of the matrix is a terminal payoff or the value of a later state. When the transitions form a DAG, the later values substitute backward. Since $V = P(\text{win}) - P(\text{loss})$, $P(\text{win}) = (1+V)/2$.

2 The game

Definition 2.1 (State)**.** *A player's profile is $z = (s, t)$: the current ST and TTD. A live state is the ordered pair $x = (z_c, z_d) = (s_c, t_c, s_d, t_d)$ from the view of the current Dropper, with $s_c, s_d \in \{0, \ldots, 299\}$ and $t_c, t_d \in \{0, \ldots, 300\}$. The terminal payoff is $+1$ if the current Dropper wins and -1 if they lose. Values are from the view of the current Dropper and negate when the roles swap.*

Definition 2.2 (Actions and timing)**.** *Both players choose seconds $d, c \in \{1, \ldots, 60\}$ at the same time (drop and check). The check succeeds if and only if $d \le c$. Then the Checker's ST grows by the inclusive interval $\ell = c - d + 1 \in \{1, \ldots, 60\}$; equal seconds cost one second, not zero. If $s_c + \ell \ge 300$, the cylinder overflows and the Dropper gets payoff $+1$. If not, play continues with the roles swapped:*

$$(s_c, t_c, s_d, t_d) \xrightarrow{\textit{success, lag } \ell} (s_d,\ t_d,\ s_c + \ell,\ t_c).$$

Definition 2.3 (Failure, dose, and revival)**.** *On a failed check ($d > c$) the Checker gets the dose $q = s_c + 60$. Survival is possible if and only if*

$$q < 300 \ \wedge\ q + t_c \le 300 \qquad \Longleftrightarrow \qquad s_c \le 239 \ \wedge\ s_c + t_c \le 240. \tag{1}$$

A profile is revival-eligible when (1) holds and failure-fatal when it does not. Both terms describe the next failed check; both players are alive in a live state. A revival-eligible Checker revives with probability

$$p(s_c, t_c) \ = \ 0.95 \left(1 - \frac{s_c}{240}\right) 0.75^{\,t_c/60}. \tag{2}$$

Then the cylinder resets and the TTD absorbs the dose:

$$(s_c, t_c, s_d, t_d) \xrightarrow{\textit{failure, revive}} (s_d,\ t_d,\ 0,\ t_c + s_c + 60) \quad \textit{with probability } p(s_c, t_c).$$

If not, the transition ends with payoff $+1$. When (1) fails, $p \equiv 0$.

The source fixes the shape of (2): the vial plus sixty seconds against a five-minute cap, and revival prospects that decrease with accumulated deaths [1]. It does not fix linearity, the product form, or the constants 0.95 and 0.75; those are choices of the model. A coarse six-action sensitivity check moved the baseline 0.95 alone across 0.85–0.99. The root value moved by 0.0028, and the Dropper's first-action mass moved from 0.387 to 0.257. The value is stable to that parameter; the policy is not.

3 Finiteness and stage structure

Lemma 3.1 (Damage increases)**.** *Let $R(x) = s_c + t_c + s_d + t_d$. Every live transition increases R. A successful lag ℓ adds $\ell \ge 1$. A revived failure adds exactly 60.*

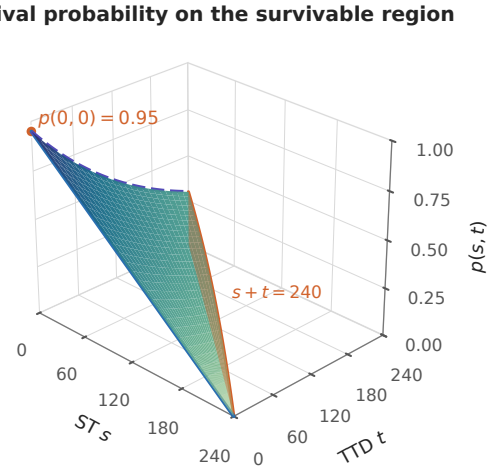


Figure 1: The frozen revival law (2). Color and height both show probability. The blue edge is the linear loss from ST. The purple edge is the exponential loss from TTD. The pale plateau is the zero-probability region, which contains $(240, 240)$. The orange curtain marks $s + t = 240$: probability is positive on the curtain, except at $s = 240$, and is zero beyond it.

Corollary 3.2 (The game is finite)**.** *$R \leq 1198$ on live states, so every play ends within 1199 transitions. Thus every live state has a unique value $V(x) \in [-1, 1]$. Backward induction over the finite acyclic transition graph gives it [2], with von Neumann's minimax theorem at each state [3]. The construction is the finite, absorbing case of a Shapley stochastic game [4].*

Proposition 3.3 (The 61 transition classes)**.** *Fix a live state x and let V give the values of its children. Write the Dropper's continuation payoff after a success of lag ℓ, or after a failure, as*

$$S_\ell = \begin{cases} +1, & s_c + \ell \geq 300, \\ -V(s_d, t_d, s_c + \ell, t_c), & \textit{else}, \end{cases}$$

$$F = \begin{cases} +1, & (s_c, t_c) \textit{ failure-fatal}, \\ p\left(-V(s_d, t_d, 0, t_c + s_c + 60)\right) + (1 - p), & \textit{else}. \end{cases}$$

Then the stage matrix of x, with the Dropper as the row player, is

$$M(x)_{d,c} = \begin{cases} S_{c-d+1}, & c \geq d, \\ F, & c < d, \end{cases} \tag{3}$$

and $V(x) = \mathrm{val}(M(x))$. $M(x)$ has at most 61 different entries: it is Toeplitz on and above the diagonal and constant below it.

Proof. A successful cell depends on the actions only through the lag $\ell = c - d + 1$. Every failed cell has the same transition distribution. The entries are the expected continuation values, negated for the role swap. □

Lemma 3.4 (Constant-time saddle test)**.** *Row d of (3) contains the prefix $S_1, \ldots, S_{61-d}$, and also F when $d > 1$. Column c contains the prefix $S_1, \ldots, S_c$, and also F when $c < 60$. Let $m = \min_\ell S_\ell$*

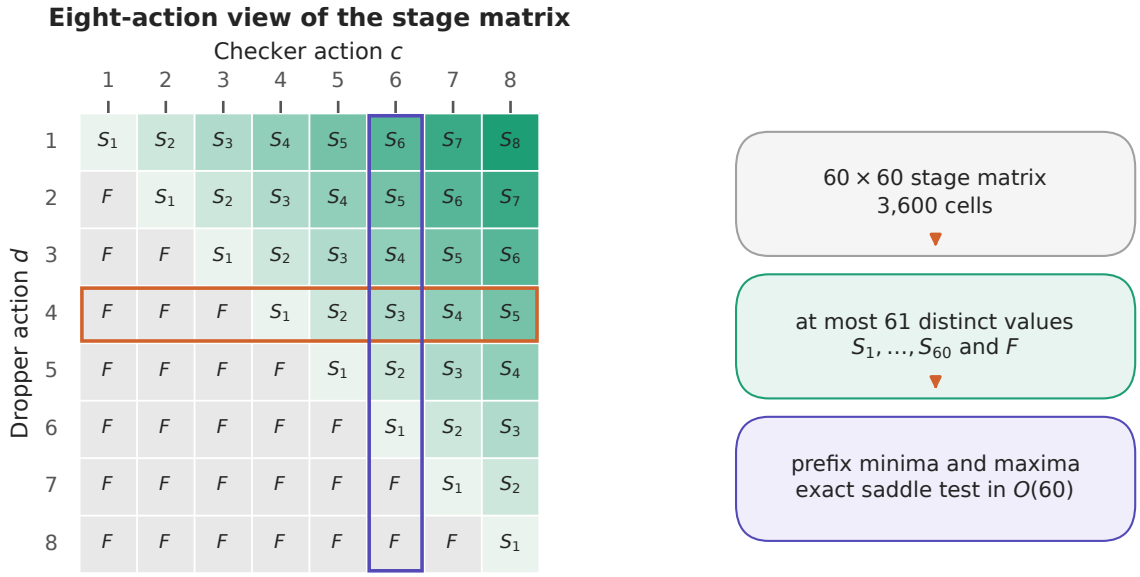


Figure 2: An 8×8 view of the stage matrix (3). The lag $\ell = c - d + 1$ is constant on each diagonal. Every failed check has the value F. Row minima and column maxima reduce to prefix scans of S (Lemma 3.4), with the same result as a scan of all 3,600 cells.

and $\overline{m} = \max_\ell S_\ell$. *Then*

$$\max_d \min_c M(x)_{d,c} = \max\big(m,\ \min(S_1, F)\big), \qquad \min_c \max_d M(x)_{d,c} = \min\big(\overline{m},\ \max(S_1, F)\big).$$

Among rows $d > 1$, *the row minimum is largest at* $d = 60$. *Among columns* $c < 60$, *the column maximum is smallest at* $c = 1$. *The test is exact, because min and max select one of their arguments and add no rounding.*

Figure 2 shows the layout. Each row contains a prefix of the S array, never an arbitrary subset. Thus the minimum and the maximum of the 60 success values, plus two comparisons, give every row minimum and column maximum. The matrix is never built.

4 The failure-fatal quotient

The model reads a player's TTD only in the revival probability (2). After failure becomes fatal, the exact TTD has no more effect. Profiles $(250, 60)$ and $(250, 300)$ both get dose 310 on a failed check, and success only raises ST. Thus they can share the sentinel $(250, \perp)$.

Lemma 4.1 (Failure-fatality is absorbing)**.** *If* (s, t) *is failure-fatal, then* (s', t) *is failure-fatal for every* $s' \geq s$. *A successful check raises ST and keeps TTD. A failed check ends the game for that player, or needs* (1). *Thus no transition makes the profile revival-eligible again.*

Theorem 4.2 (Quotient soundness)**.** *Let* $\varphi(s, t) = (s, t)$ *when* (s, t) *is revival-eligible and* $\varphi(s, t) = (s, \perp)$ *when it is not, applied to each player's profile independently. Two live states with equal images under* φ *have identical stage matrices up to* φ*-identification of their children. Thus they have identical values:* V *factors through* φ.

Proof. Compare the 61 classes of Proposition 3.3 at two states that differ only in a failure-fatal player's TTD. Success classes do not read TTD. A child's φ-image depends only on the parent's

image and ℓ (Lemma 4.1). The failure class reads the Checker's TTD only through p. A failure-fatal Checker gives $F = +1$ with no child, and a revival-eligible Checker is not quotiented. Induction over the finite acyclic graph completes the proof. □

Proposition 4.3 (Counts and closure)**.** *The root TTDs lie in* $\{0\} \cup [60, 300]$*, and the domain is closed under transitions. Revival needs* $s_c + t_c \leq 240$*, so a failure child's TTD* $t_c + s_c + 60$ *lies in* $[60, 300]$*; success keeps TTDs. Over this domain, one player has* $300 \times 242 = 72{,}600$ *profiles. The quotient keeps all* 16,711 *revival-eligible profiles and replaces the failure-fatal TTDs at each ST with one sentinel. This leaves* 17,011 *classes per player and*

$$17{,}011^2 = 289{,}374{,}121$$

two-player classes, against $72{,}600^2 = 5{,}270{,}760{,}000$ *before the quotient. A revival-eligible profile with TTD in* 1..59 *is unreachable, and the implementation rejects it.*

Figure 3 shows why the reduction is large. Revival-eligible profiles are 23% of the one-player domain. Both profiles are revival-eligible in only 5.3% of states.

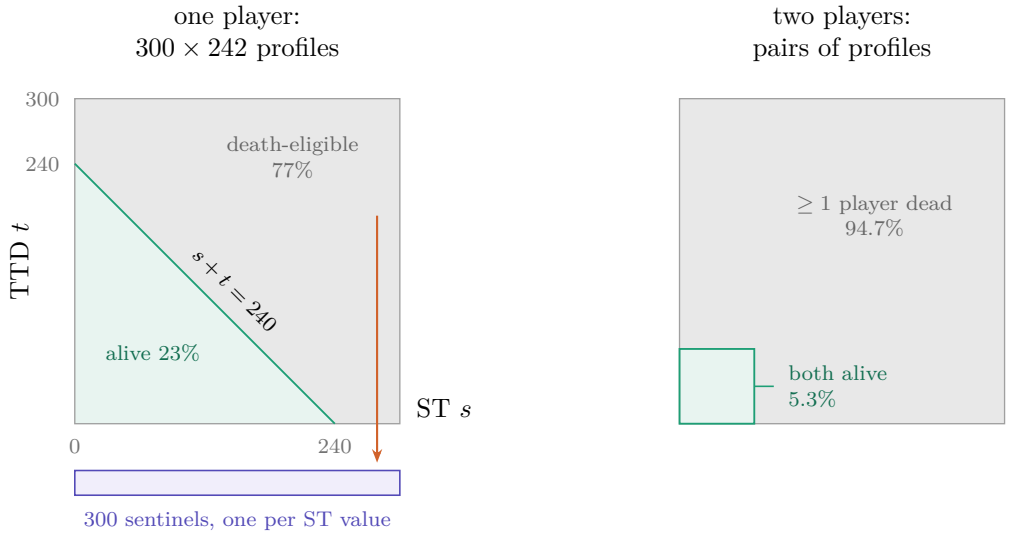


Figure 3: The quotient for one player (left) and for pairs of players (right). Each vertical fibre in the death-eligible region collapses to one sentinel per ST value. The left panel is schematic, because TTD values $1, \ldots, 59$ are unreachable. The percentages are exact counts on the closed domain. "Death-eligible" and "dead" mean failure-fatal. Neither means that a player has left the game.

5 The potential and the sweep schedule

The potential is not payoff, value, or strategic advantage. It is only a progress score that orders the states.

Definition 5.1 (Potential)**.** *For a profile let* $\rho(s, t) = t$ *if* (s, t) *is revival-eligible and* $\rho(s, t) = 301$ *if it is not. For a class let*

$$\Phi(x) = s_c + s_d + \rho(s_c, t_c) + \rho(s_d, t_d) \in \{0, \ldots, 1200\}.$$

Theorem 5.2 (Strict increase)**.** *Every live transition increases* Φ*. The cases for the profile that moves are:*

- *successful lag ℓ, same side of the revival boundary:* $\Delta\Phi = \ell \geq 1$;
- *successful lag ℓ, the profile becomes failure-fatal:* $\Delta\Phi = \ell + 301 - t \geq \ell + 61$, *because revival-eligibility implies* $t \leq 240$;
- *failure, the revived profile stays revival-eligible:* $\Delta\Phi = 60$, *because ST resets while TTD absorbs the dose;*
- *failure, the revived profile becomes failure-fatal:* $\Delta\Phi = 301 - (s_c + t_c) \geq 61$ *by* (1).

Corollary 5.3 (Layered backward induction)**.** *No transition stays inside a Φ-layer. Thus a descending sweep solves every child before its parents. With the one-player bucket $B_a = \{(s,t) : s + \rho(s,t) = a\}$, layer P is exactly*

$$\Phi^{-1}(P) = \bigcup_{\substack{0 \leq a \leq 600 \\ 0 \leq P-a \leq 600}} B_a \times B_{P-a}.$$

The 289,374,121 *classes fill all* 1,201 *layers. The largest layer,* $\Phi = 374$, *holds* 1,678,715 *classes.*

The classes form the layered DAG in Figure 4. The table build asserts the increase over every profile transition. The schedule needs no reachability analysis, frontier queue, or transposition table, and it solves unreachable classes too.

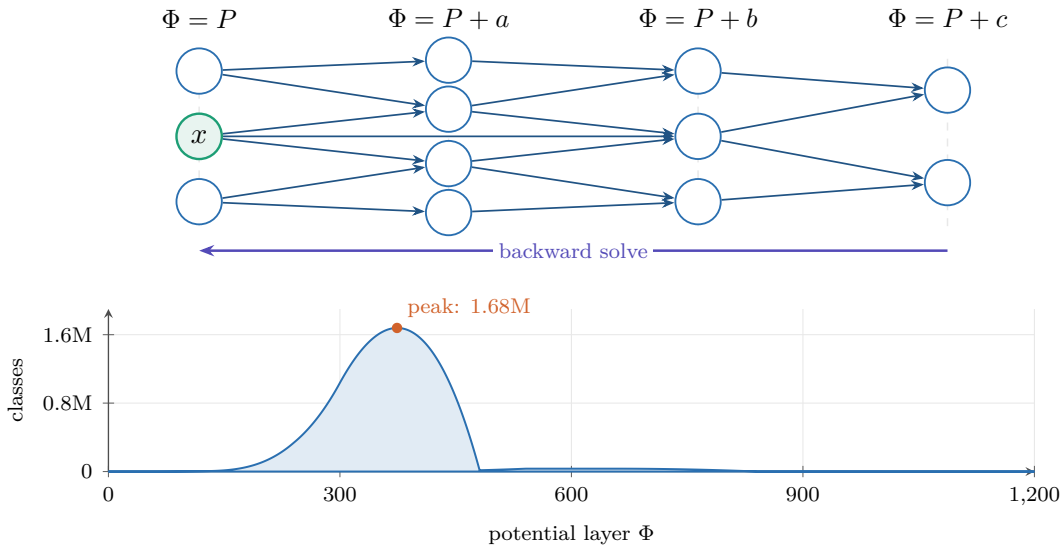


Figure 4: Top: some transitions in the potential-ranked DAG. Bottom: the exact width of all 1,201 layers on a linear scale. Every edge points right; the computation runs from right to left.

6 Certified equilibria

Definition 6.1 (Certificate)**.** *For a stage matrix M and mixed strategies σ_d (Dropper) and σ_c (Checker), let*

$$L = \min_c (\sigma_d^{\mathsf{T}} M)_c, \qquad U = \max_d (M\sigma_c)_d, \qquad \mathrm{gap}_M(\sigma_d, \sigma_c) = \max(0,\, U - L).$$

The pair certifies the value $\widehat{v} = (L+U)/2$ at tolerance τ_0 when $\mathrm{gap}_M(\sigma_d, \sigma_c) \leq \tau_0$. In this paper $\tau_0 = 10^{-6}$. The gap is always measured against the full matrix (3).

Proposition 6.2 (Certified midpoint)**.** *$L \le \mathrm{val}(M) \le U$ for every mixed pair. Thus a certificate at tolerance τ_0 gives $|\widehat{v} - \mathrm{val}(M)| \le \tau_0/2$. If $\max_{d,c} |M_{d,c} - \widehat{M}_{d,c}| \le \epsilon$, then $|\mathrm{val}(M) - \mathrm{val}(\widehat{M})| \le \epsilon$.*

Proposition 6.3 (One recurrence, both strategies)**.** *Let $d_0 = S_1 - F$ and $\Delta_m = S_{m+1} - S_m$. If $d_0 \neq 0$, define*

$$r_1 = 1, \qquad r_k = -\frac{1}{d_0} \sum_{m=1}^{k-1} \Delta_m \, r_{k-m} \quad (k = 2, \ldots, 60).$$

Then $\sigma_d \propto (r_1, \ldots, r_{60})$ makes $\sigma_d^{\mathsf{T}} M$ constant, $\sigma_c \propto (r_{60}, \ldots, r_1)$ makes $M\sigma_c$ constant, and $(M\sigma_c)_i = (\sigma_d^{\mathsf{T}} M)_{61-i}$ for every i.

Proof. Row $d+1$ of (3) is row d moved one column to the right, with F in column d. Thus, for $d = 1, \ldots, 59$,

$$(M\sigma_c)_d - (M\sigma_c)_{d+1} = d_0 \, \sigma_{c,d} + \sum_{m \ge 1} \Delta_m \, \sigma_{c,d+m}.$$

This is zero when the Dropper is indifferent between seconds d and $d+1$. Zero gives $\sigma_{c,d}$ from the later entries. Start at $\sigma_{c,60} = 1$ and step down 59 times: this is the recurrence with $r_k = \sigma_{c,61-k}$. The same subtraction on columns c and $c+1$ gives $d_0 \, \sigma_{d,c+1} + \sum_{m \ge 1} \Delta_m \, \sigma_{d,c+1-m}$, the same coefficients read forwards from $\sigma_{d,1} = 1$, so $\sigma_{d,k} = r_k$. Substitute $k \mapsto 61 - k$ in the two sums to get $(M\sigma_c)_i = (\sigma_d^{\mathsf{T}} M)_{61-i}$. □

Negative entries of r are clipped, and the pair is normalized, before the certificate is measured. The clip cannot invalidate the bounds, because they hold for any mixed pair. Each class then goes through the same ladder:

1. **Pure saddle.** Lemma 3.4. If the maximin and the minimax meet within τ_0, a pure action pair is enough.
2. **Equalizer.** Proposition 6.3: 1,770 multiply-adds for r and 1,770 for $M\sigma_c$. Its maximum and minimum are U and L.
3. **Linear programming.** The standard zero-sum program $\max\{v : M^{\mathsf{T}}\sigma_d \ge v\mathbf{1},\ \mathbf{1}^{\mathsf{T}}\sigma_d = 1,\ \sigma_d \ge 0\}$, with σ_c from its duals, then the same full-matrix check.

Proposition 6.4 (A priori accumulated bound)**.** *Let every stored value have a certificate at tolerance τ_0. Let $\delta(x)$ bound the longest live path from x to a terminal. Then*

$$|\widehat{V}(x) - V(x)| \;\le\; \delta(x)\,\frac{\tau_0}{2} \;\le\; \frac{1201 - \Phi(x)}{2}\,\tau_0,$$

so the root state obeys $|\widehat{V} - V|(0,0,0,0) \le 6.1 \times 10^{-4}$.

Proof. Induction up the DAG. Terminal entries are exact. If every child is within $(\delta(x) - 1)\tau_0/2$ of truth, then the assembled matrix is within that distance entrywise, because expectation and negation are non-expansive. Proposition 6.2 moves the bound to the value and adds the state's own $\tau_0/2$. Theorem 5.2 bounds δ. □

Proposition 6.5 (Full-policy bound)**.** *Let $\widehat{\pi}$ play the accepted pair at every class. Let $\underline{V}_{\widehat{\pi}}(x)$ and $\overline{V}_{\widehat{\pi}}(x)$ be its security values against every complete opposing policy, with deviations after later role swaps included. Then*

$$\overline{V}_{\widehat{\pi}}(x) - \underline{V}_{\widehat{\pi}}(x) \le \varepsilon(x) \le (1201 - \Phi(x))\tau_0,$$

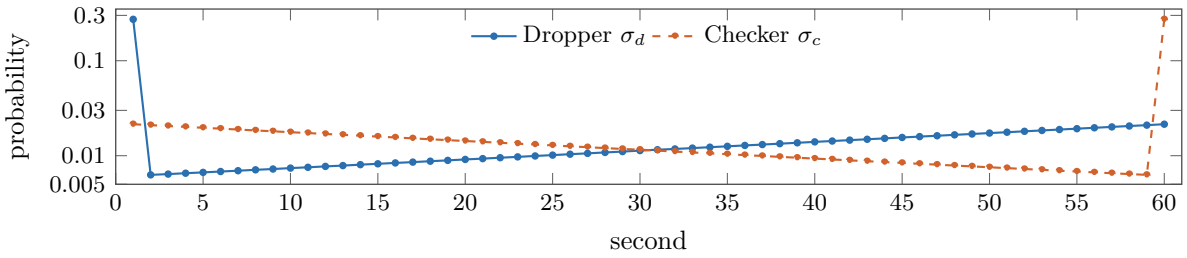


Figure 5: The first-round strategies of the selected full-game policy, both with full support. The Dropper puts 27.3% on second 1, and a ramp from 0.6% to 2.2% on the other seconds. The Checker is the exact mirror $\sigma_{c,j} = \sigma_{d,61-j}$ of Proposition 6.3. Proposition 6.5 shows how later gaps accumulate.

where $\varepsilon(x) = (U_x - L_x) + \max_y \varepsilon(y)$ *over live children* y*, and* $\varepsilon = 0$ *at terminals. At the root state, no unilateral deviation improves the win probability by more than* 6.005×10^{-4}*.*

Proof. At a terminal, both security values equal $\widehat{V}$. Assume the two bounds for every live child y: $\underline{V}_{\widehat{\pi}}(y) \geq \widehat{V}(y) - \frac{1}{2}\varepsilon(y)$ and $\overline{V}_{\widehat{\pi}}(y) \leq \widehat{V}(y) + \frac{1}{2}\varepsilon(y)$. Negation at the role swap and the average after revival keep the largest child bound. Thus $\underline{V}_{\widehat{\pi}}(x) \geq L_x - \frac{1}{2}\max_y \varepsilon(y)$ and $\overline{V}_{\widehat{\pi}}(x) \leq U_x + \frac{1}{2}\max_y \varepsilon(y)$. With $\widehat{V}(x) = (L_x + U_x)/2$, this is the recurrence for ε. □

7 The computation

The implementation is one Python file of 476 lines. It builds the 17,011 quotient profiles with their successful-check children, failure child, and revival probability. It stores V as a $17{,}011 \times 17{,}012$ float64 array. The extra column holds -1, so a terminal win reads as $+1$ after the role-swap negation, with no branch. It lists each layer by Corollary 5.3, in an order where consecutive classes read the same row of V. One compiled kernel per layer runs in parallel over the classes.

For each class, the kernel gathers the 61 continuation values. It runs Lemma 3.4, then Proposition 6.3, in sixty-entry scratch that stays in first-level cache. It tracks the minimum and the maximum of $M\sigma_c$ as they are produced. It writes the certified midpoint and its rung. Classes that fail both rungs are collected and solved by linear programming afterwards.

7.1 Results

The sweep covered all 289,374,121 classes in 48.9 seconds on an Apple M5 Pro laptop. The laptop has five performance cores and ten efficiency cores; the kernel used fifteen threads with a static schedule. That is about 5.9 million classes per second, and a single thread does 0.49 million per second. The earlier artifact came from the same rules on a twelve-core desktop: 5,432 seconds for 98.1% of the table. It used a warm-started support search, two bordered 61×61 linear systems per class, and sessions with checkpoints.

Three changes explain the ratio. Proposition 6.3 replaces about 1.6×10^5 floating-point operations per class with about 3.6×10^3. The fused kernel removes every temporary array. The clip of the equalizer, in place of a rejection, sends no class to the linear program.

rung	classes	share
pure saddle (Lemma 3.4)	334,177	0.115%
equalizer (Proposition 6.3)	289,039,944	99.885%
linear programming	0	0

The earlier artifact sent 190,995 classes to linear programming and 334,588 to the pure rung. The difference is borderline classes with a gap at the tolerance; both routes are certified.

Six stored values were compared with the earlier artifact. Five agree to within 2×10^{-11}; the state $(250, 300, 40, 0)$ agrees to 1.2×10^{-9}. After the sweep, an independent scalar ladder re-solved 1,200 evenly strided classes from their stored children. The largest difference from the stored value was 8.9×10^{-16}. The root's local gap is 5.6×10^{-17}. Figure 5 shows its strategies.

Theorem 7.1 (Main result)**.** *The complete 60-action timing model of Section 2 has a certified numerical solution on the closed quotient domain of Proposition 4.3. Every one of its* 289,374,121 *classes has a stored midpoint with local gap at most* 10^{-6}*. At the root state the stored midpoint is* 0.0898500728141386*, and*

$$V(0,0,0,0) = 0.08985 \pm 0.00061, \qquad P(\textit{first Dropper wins}) = 0.54493 \pm 0.00031.$$

For the selected full-game policy, $\overline{V}_{\widehat{\pi}}(0,0,0,0) - \underline{V}_{\widehat{\pi}}(0,0,0,0) \leq 1.201 \times 10^{-3}$.

8 Scope of the claims

- **Rules-relative.** The results concern Section 2 and the frozen revival surface (2). They do not concern the leap-aware game with its asymmetric 61st second.
- **Certified numerics.** Values are round-to-nearest float64 midpoints with saddle-gap certificates, not rational or interval arithmetic. The structural theorems are exact. The table build asserts survival, absorption, the counts, and the potential increase by enumeration.
- **Domain.** States with a revival-eligible profile with TTD in 1..59 are outside the table; the implementation rejects them.
- **Policies.** The table stores values and routes; Proposition 6.3 re-derives strategies from child values in microseconds.

Reproducibility. From the repository root, `uv run src/dth.py` builds the table and prints the six reference values in under a minute. `uv run pytest` runs the nine structural and numerical checks.

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